Research Log Book

20 March - 10 October 2026

AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing

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Individual component	Cloth Pieces Counting and Operator Cycle Time Analysis

March 2026

Research work record

20-03-2026	* After the proposal presentation, I reviewed the feedback related to my feature. * I understood that my feature must count cloth pieces correctly and measure the time taken by the operator to complete one piece.
23-03-2026	* I identified that my feature mainly works in two places: the sewing operator area and the quality checker area.
24-03-2026	* I first focused on the sewing machine operator area. * I studied how to monitor the operator's work and how to identify completed garment pieces.
25-03-2026	* I worked on the idea of counting how many garment pieces are sewn by the operator. * I planned that the system should detect each completed garment piece correctly.

March 2026

Research work record

26-03-2026	* I studied how to calculate the time taken by the operator to complete one garment piece. * I planned that the system should detect each completed garment piece correctly.
27-03-2026	* I decided that I needed to observe the actual garment production process before finalising the feature logic.
28-03-2026	* I took part in a factory visit to understand the real working process. * The visit helped me connect my feature idea with the actual production line.

March 2026

Research work record

29-03-2026 to 30-03-2026	* I visited MAS Sleekline, Nittambuwa, and observed how garment production happens in the factory. * I paid special attention to how operators handle garment pieces during sewing. * During the factory visit, I observed that around 8-10 workers are involved in completing one garment piece. * This helped me understand that one garment passes through several operators before it is fully completed.
31-03-2026	* I carefully observed how garment pieces are passed from one operator to another in the production line. * I noticed that there is a fixed working flow and passing pattern between operators.

April 2026

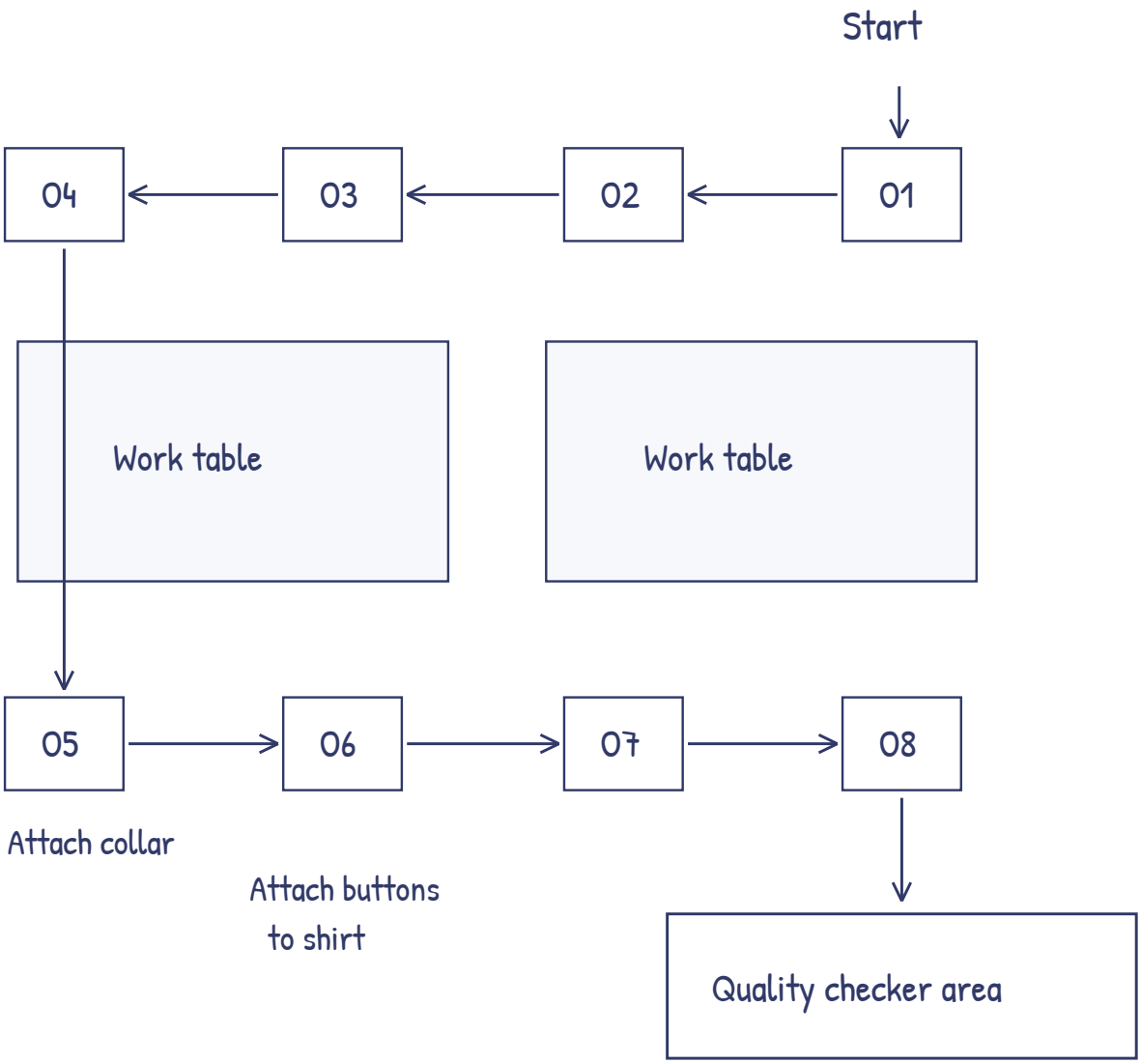
Research work record

01-04-2026 to 03-04-2026	* I further studied the movement pattern of garment pieces in the production line. * I noticed that garment pieces move in zig-zag and linear patterns between workers until the full garment is completed. * After observing that line, I realised that my system should focus on one specific operator rather than try to monitor the whole line at once. * I decided that my feature should capture one selected sewing operator and monitor only that operator's work.

Production-Line Observation

Sketch linked to 01-04-2026 to 03-04-2026

Garment pieces pass through several operators before reaching the quality checker. This diagram redraws the movement pattern recorded in the original logbook.



01-08: operators in the observed production line.

The observation led to a narrower scope: monitor one selected operator and that operator's output area.

April 2026

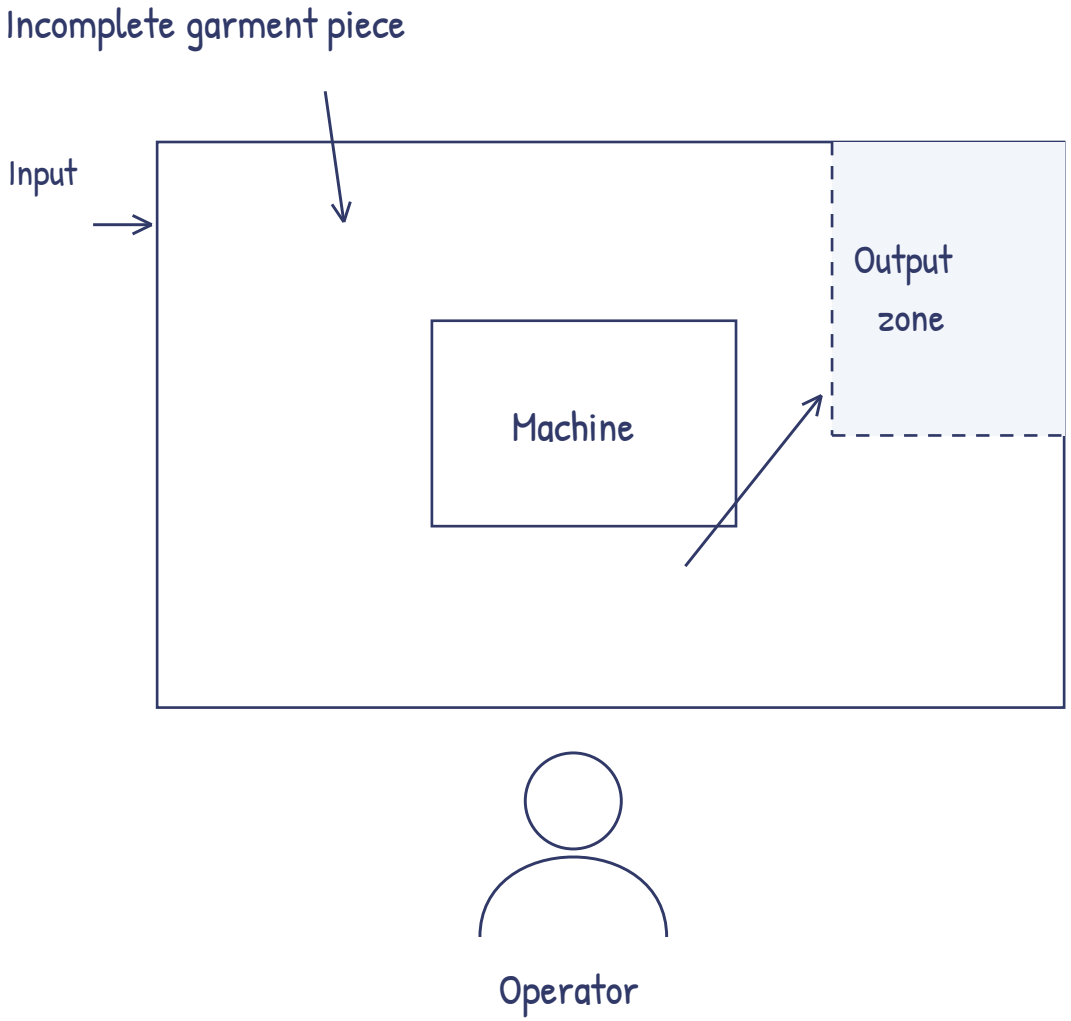
Research work record

04-04-2026	* I studied where the selected operator places the garment piece after the sewing task. From this observation, I realised that the system must identify or configure a specific output zone to detect completed pieces correctly. * I further improved this output-zone idea. * I understood that the output zone is the most important area for counting garment pieces and measuring operator cycle time.

Selected Operator and Output Zone

Sketch linked to 04-04-2026

The early design considered a specific output zone for detecting the garment piece placed aside after sewing.



The operator places the finished garment in the output zone after completing the sewing task.

Example workstation: collar or button operation for a shirt.

April 2026

Research work record

05-04-2026 to 06-04-2026	* After the factory observation, I discussed the need for actual video footage for my feature. * I identified that video footage is necessary to study operator movement and output placement more accurately.
07-04-2026	* We requested video footage from the factory for our features. * The purpose was to analyse the selected operators' working process in more detail and train an ML model.
08-04-2026	* I reviewed the feature flow again using the observations gathered from the factory. * I improved the planned sequence as observing the operator, detecting the completed piece and checking the output location.
09-04-2026	* I continued working on the logic for counting garment pieces from the selected operator. * I focused on making sure the system counts only after the garment enters the output zone.

April 2026

Research work record

10-04-2026 to 14-04-2026	* I also improved the idea for operator cycle-time calculation. * I planned to record the completion time of one garment piece and compare it with the next completed piece. * I studied how the rework process affects my feature. * I studied how the downtime process affects my feature.
17-04-2026	* I received the requested video footage from the factory around this period. * I started checking whether the footage could support the selected operator monitoring and output-zone analysis. * I reviewed the received footage against my feature idea.
18-04-2026	* I finalised my progress for this period. * By this stage, I had completed the feature understanding through early research, factory observation, identification of the selected operator and output zone, and collection of video footage for the next step.

April 2026

Research work record

19-04-2026	* After receiving the video footage, I started studying it carefully. * I checked how the operator works, where the garment piece is sewn and where the finished piece is placed after sewing.
20-04-2026	* At first, I thought the system could be built by fixing the camera and defining the output zone clearly. * My idea was that when the operator puts the finished garment piece into the output zone, the counter should increase.
21-04-2026	* After reviewing the factory process again, I understood that every operator has a separate output area. * This helped me understand that the system must be designed for one specific operator and one specific output zone.
22-04-2026	* Sometimes the output zone is overlapped by the garment piece being sewn, and this can cause incorrect detection and piece counting.

April 2026

Research work record

23-04-2026 to 24-04-2026	* I analysed how the camera view would look when fixed above the sewing operator. * From the top view, the camera can capture the machine, the operator's hand movements, the garment piece being sewn, the output zone and sometimes even the operator's head and shoulders.
25-04-2026	* I identified that these extra visible parts can affect the system performance. * I understood that the model must be trained using real factory conditions instead of only clean video samples.
26-04-2026	* I started planning how to organise the raw video footage before annotation. * I understood that each video must be checked properly before using it for training.
27-04-2026	* I worked on the video audit and triage plan for the dataset. * I decided to log important details such as video duration, FPS, camera angle, visible body parts, lighting conditions, output zone and quality issues.

April 2026

Research work record

28-04-2026 to 30-04-2026	* I continued improving the dataset preparation idea. * I studied whether the messy videos should be removed or kept. * I decided to prepare the dataset using frame extraction for object detection and short clips for action recognition.

May 2026

Research work record

01-05-2026	* I continued the video audit started in April and arranged the raw recordings in a clear folder structure. * I kept the original videos separate from files prepared for training.
04-05-2026	* I checked video duration, frame rate and camera position before annotation. * I noted recordings where the sewing area or cloth movement was difficult to see.
05-05-2026	* I reviewed examples from the sleeves, collar and pocket operations. * I focused on the work of one selected operator rather than counting the whole production line.
06-05-2026	* I examined how the operator prepared cloth, stitched it and moved it away. * I used these observations to define the action labels consistently.

May 2026

Research work record

07-05-2026	* I prepared the segment annotation format with video ID, start time, end time and event type. * I checked how to record changes between activities using video timestamps.
08-05-2026	* I started marking IDLE_SETUP and SEWING intervals in the selected recordings. * I reviewed unclear boundaries instead of assigning labels only from a single frame.
11-05-2026	* I continued annotating the recordings using the same state definitions. * I paid attention to cloth adjustment so it would not be confused with active sewing.
12-05-2026	* I examined PUT_ASIDE examples as a possible signal of a completed garment. * I noted that this action could be brief and difficult to separate from normal cloth handling.

May 2026

Research work record

13-05-2026	* I reviewed pauses in the recordings and their effect on the state labels. * I kept the main research focus on production actions and completion events.
14-05-2026	* I checked the annotation table for missing values and incorrect time intervals. * I corrected rows where an end time did not match the visible activity boundary.
15-05-2026	* I studied how to convert annotated time segments into short training clips. * I selected a fixed clip window so the model could learn from recent motion.
18-05-2026	* I prepared the short-clip extraction workflow using the annotated segments. * I checked sample clips against their source videos to confirm the assigned actions.

May 2026

Research work record

19-05-2026	* I organised the video-level training, validation and test groups. * I kept clips from one source video in the same group to reduce data leakage.
20-05-2026	* I reviewed the number of examples available for each action class. * I noted how short or rare actions could affect training and evaluation.
21-05-2026	* I studied the earlier CNN window-based approach as an initial model direction. * I compared its full-video processing flow with the need for continuous live updates.
22-05-2026	* I examined the delay caused by generating and processing many overlapping windows. * I identified the need for a lighter model that could work on a rolling clip.

May 2026

Research work record

25-05-2026	* I reviewed how state probabilities should become garment completion events. * I decided that a single prediction should not directly increase the garment count.
26-05-2026	* I prepared the basic event-decoder design using activity order and timestamps. * I considered stability rules to reduce repeated counts from the same action.
27-05-2026	* I reviewed lightweight model options for recognising short sewing activities. * I focused on MobileNetV3-Small as a practical feature extractor for the next experiment.
29-05-2026	* I organised the prepared annotations, clip workflow and model-development notes. * I planned the next stage around temporal classification and event-based counting.

June 2026

Research work record

01-06-2026	* I continued developing the temporal activity classifier using MobileNetV3-Small. * I used several frames from each short clip so the model could learn activity changes.
02-06-2026	* I prepared the frame resizing and normalisation steps used by the model. * I checked that the input format stayed consistent across training and evaluation.
03-06-2026	* I checked the dataset loader and the labels returned for each sampled clip. * I reviewed example batches before starting longer training runs.
04-06-2026	* I worked on the temporal model training pipeline and saved model checkpoints. * I kept training and validation outputs separate for later review.

June 2026

Research work record

05-06-2026	* I reviewed the training progress and compared validation predictions with the labels. * I looked for differences between the common actions and the short PUT_ASIDE action.
08-06-2026	* I examined incorrect predictions around the end of sewing and the start of cloth handling. * I found that PUT_ASIDE was less dependable as the main counting signal.
09-06-2026	* I compared a three-state interpretation with a simpler two-state approach. * I focused on whether IDLE_SETUP and SEWING could describe the useful production cycle.
10-06-2026	* I revised the action-classification workflow to use IDLE_SETUP and SEWING. * I removed PUT_ASIDE from the final counting rule because its predictions were unreliable.

June 2026

Research work record

11-06-2026	* I prepared the two-state annotations and checked their label mapping. * I kept the same meaning of each state when preparing model inputs and reports.
12-06-2026	* I trained and reviewed the two-state temporal MobileNetV3-Small model. * I compared the predicted activities with the sequence visible in the test videos.
15-06-2026	* I developed the rule that a confirmed SEWING state followed by IDLE_SETUP represents one piece. * I made sure continuous predictions of the same state would not create extra pieces.
16-06-2026	* I worked on temporal smoothing and stable-state confirmation in the decoder. * I reviewed rapid state changes that could otherwise cause false garment counts.

June 2026

Research work record

17-06-2026	* I calculated cycle time from successive accepted IDLE_SETUP cycle events. * I checked that time values came from the video timeline during recorded-video evaluation.
18-06-2026	* I prepared manual completion-event timestamps for comparison with decoded events. * I kept state-classification evaluation separate from count and cycle-time evaluation.
19-06-2026	* I compared predicted garment counts with the manual counts in selected recordings. * I reviewed missing and extra events rather than relying only on classification accuracy.
22-06-2026	* I reviewed cycle-time differences and the timing of accepted events. * I saved the evaluation outputs so the results could be explained later.

June 2026

Research work record

24-06-2026	* I studied how a workstation detector could restrict classification to the sewing area. * I planned a workstation region of interest to reduce the effect of unrelated background content.
25-06-2026	* I prepared video frames for workstation annotation and organised the image output. * I used the workstation class consistently when marking the sewing area.
29-06-2026	* I reviewed the frame-extraction and frame-shuffling scripts for the detector dataset. * I combined the workflow into extract_and_shuffle_frames.py with shuffled_frames as the output folder.
30-06-2026	* I worked on the YOLOv8 workstation-detection training workflow and reviewed its output. * I prepared the detector and temporal classifier for a combined vision pipeline.

July 2026

Research work record

01-07-2026	* I organised the model-development outputs and the two-state event-decoder workflow. * I reviewed the path from a short video clip to a piece event and cycle-time value.
02-07-2026	* I reviewed the phase results and the prepared annotation and evaluation records. * I separated model classification results from production-count and cycle-time results.
03-07-2026	* I worked on passing the detected workstation region into the temporal classifier. * I checked that the crop and image preprocessing matched the classifier input format.
06-07-2026	* I prepared the first application flow using FastAPI and a web dashboard. * I planned how the backend would send the current state, piece count and cycle time.

July 2026

Research work record

07-07-2026	* I worked on recorded-video input for testing the full inference pipeline. * I read the recording as a stream so results could update while the video played.
08-07-2026	* I connected the accepted state transitions to the application count display. * I checked that one completion event produced one update in the session records.
09-07-2026	* I worked on displaying the latest cycle time and the session average. * I checked how the application should handle the period before a complete cycle was available.
10-07-2026	* I prepared session-based storage for garment counts and event timestamps. * I kept each session's production data separate so history could be reviewed later.

July 2026

Research work record

13-07-2026	* I worked on the create-session flow with employee selection and a target piece count. * I linked the selected employee with the relevant sewing-line information.
14-07-2026	* I reviewed the live dashboard arrangement for the camera and session information. * I aimed to keep the important production values visible during monitoring.
15-07-2026	* I improved the cycle-time trend display for individual completed pieces. * I used a consistent bar width so the chart remained readable as the count increased.
16-07-2026	* I worked on the session summary and the latest cycle-time information. * I reviewed the camera and summary-card sizes to make the dashboard easier to use.

July 2026

Research work record

17-07-2026	* I worked on session history so completed sessions could be inspected separately. * I reviewed which details should appear in a session report and its expanded view.
20-07-2026	* I checked the rule that only one production session should be active at a time. * I reviewed the message shown when another session was already running.
21-07-2026	* I worked on keeping the camera under manual control in the live dashboard. * I avoided starting capture automatically when the user only moved between pages.
22-07-2026	* I reviewed test-video playback and model loading through the application. * I checked that the saved checkpoint was being used with the correct two-state class order.

July 2026

Research work record

23-07-2026	* I investigated the checkpoint-loading failure in the test-video flow. * I aligned the backend classifier dimensions with the saved TemporalMobileNetV3Small checkpoint.
24-07-2026	* I reviewed predictions after the checkpoint-loading correction. * I checked the complete chain from workstation crop to activity state and count update.
27-07-2026	* I refined the dashboard layout, sidebar visibility and camera preview area. * I kept the preview size steady so the interface would not shift during a session.
31-07-2026	* I reviewed the prototype's model, session and dashboard integration. * I recorded the remaining backend and desktop-workflow improvements for the next stage.

August 2026

Research work record

03-08-2026	* I reviewed the application structure for a packaged desktop version. * I worked toward an Electron interface with React, TypeScript and a local Python service.
04-08-2026	* I organised the FastAPI backend into clearer feature and service sections. * I separated session handling, production events, analytics and vision processing.
05-08-2026	* I worked on SQLite storage for the desktop application's local records. * I checked how sessions and piece events should be linked in the database.
06-08-2026	* I reviewed the employee and sewing-line data needed before a session starts. * I kept the session target as a value entered for the current production task.

August 2026

Research work record

07-08-2026	* I worked on device configuration and the order of the startup screens. * I placed camera selection and device settings before the production-monitoring flow.
10-08-2026	* I developed the session endpoints and the related request and response formats. * I reviewed validation for session creation, completion and history retrieval.
11-08-2026	* I continued the backend work for camera capture, preview and live classification. * I retained the confirmed SEWING to IDLE_SETUP counting rule in the integrated pipeline.
12-08-2026	* I worked on automatic persistence of accepted piece events and cycle times. * I checked that the session history could be rebuilt from stored production records.

August 2026

Research work record

13-08-2026	* I reviewed workstation-region handling during continuous video capture. * I considered how unstable or missing workstation detections should affect classification.
14-08-2026	* I reviewed camera suitability and the risk of classifying unrelated scenes. * I noted that the two-state classifier needed workstation checks around the model pipeline.
17-08-2026	* I worked on integrating the workstation detector with the live inference service. * I checked the connection between the detector crop, frame buffer and temporal classifier.
18-08-2026	* I reviewed the combined detector and two-state classifier in the application workflow. * I kept the model files and class configuration aligned with the backend loader.

August 2026

Research work record

19-08-2026	* I worked on the IoT event format for Rework, Downtime and Reset inputs. * I planned to record these device actions against the active production session.
20-08-2026	* I implemented the backend rule for Rework mode to pause piece-count increments. * I used Reset as the action to return to normal monitoring.
21-08-2026	* I worked on the Downtime rule so paused time could be excluded from cycle-time calculations. * I reviewed how Reset should resume the normal timing flow.
24-08-2026	* I prepared the ESP32-S3 Bluetooth connection flow in the device settings. * I kept device readiness as a requirement to check before starting a session.

August 2026

Research work record

25-08-2026	* I worked on production summaries and analytics endpoints using the stored session data. * I reviewed total pieces, cycle-time values and the information required for reports.
26-08-2026	* I prepared API examples and a Postman workflow for checking the backend. * I arranged the requests in the order needed to create, monitor and end a session.
27-08-2026	* I reviewed backend tests for session rules, stored events and API behaviour. * I used the test coverage to identify integration issues before packaging the application.
29-08-2026	* I assembled the complete backend package with its endpoint guide and test instructions. * I reviewed the next integration steps for the Electron desktop client.

September 2026

Research work record

01-09-2026	* I reviewed the operation-type enhancement for the model and session workflow. * I kept sleeves, collar and pocket as the operation categories used in the project.
02-09-2026	* I added operator_type information to the segment-annotation workflow. * I checked that the operation label could be traced back to the correct video.
03-09-2026	* I organised the enhancement experiment separately from the earlier model version. * I kept earlier outputs available so the development stages could be compared.
04-09-2026	* I reviewed video IDs and annotation consistency before preparing the enhanced dataset. * I checked that corrected timestamps were carried into the working annotation files.

September 2026

Research work record

07-09-2026	* I prepared the model-development workflow using the operation-labelled data. * I checked the video-based split before generating the corresponding training clips.
08-09-2026	* I continued the operation-type model experiment and reviewed its saved outputs. * I compared predicted activity states with the visible actions for each operation.
09-09-2026	* I reviewed count and cycle-time behaviour across sleeves, collar and pocket examples. * I considered how different operation lengths affected the interpretation of cycle times.
10-09-2026	* I added operation selection to the create-session flow. * I checked that the selected operation stayed linked with the current session's records.

September 2026

Research work record

11-09-2026	* I worked on the estimated cycle-time value for the selected operation. * I planned to display this estimate beside the measured production-cycle information.
14-09-2026	* I added cycle-time colour cues to the live dashboard. * I used red when a cycle exceeded the estimate and blue when it was below the estimate.
15-09-2026	* I reviewed the Electron application and the local FastAPI sidecar startup flow. * I checked the Python environment path used when the desktop application launched the backend.
16-09-2026	* I investigated the workstation-startup error caused by a missing backend Python executable. * I reviewed the virtual environment and the GARMENT_COUNTER_PYTHON setting used by the launcher.

September 2026

Research work record

17-09-2026	* I reviewed the backend setup with Python 3.11 and the required dependencies. * I updated the setup steps so the desktop launcher could find the intended Python environment.
18-09-2026	* I checked service startup, model-file paths and camera access after the environment changes. * I continued reviewing the connection between the desktop interface and the local API.
21-09-2026	* I reviewed the database location under the application's user-data folder. * I checked how an unfinished active session could affect later attempts to start monitoring.
22-09-2026	* I reviewed the live dashboard with the camera preview and current-session summary. * I kept the video overlay focused on the activity state while counts stayed in the dashboard.

September 2026

Research work record

23-09-2026	* I reviewed the completed-session cards and the detailed history view. * I checked how saved cycle-time values were presented when opening an earlier session.
24-09-2026	* I prepared the recorded-video demonstration workflow for the live dashboard. * I kept the test-video upload in the monitoring screen instead of the start-session form.
25-09-2026	* I reviewed the desktop run instructions and the order of the device, session and monitoring steps. * I documented remaining device checks separately from the software implementation work.
30-09-2026	* I organised the enhanced model outputs, application files and development notes. * I prepared the evidence needed for the individual research report and final demonstration.

October 2026

Research work record

01-10-2026	* I gathered the component's methodology, model outputs and application screenshots for documentation. * I kept the vision-based counting and cycle-time scope clear in the report outline.
02-10-2026	* I reviewed the model pipeline from workstation detection to temporal classification and event decoding. * I checked that the diagrams used the final two-state counting rule.
03-10-2026	* I reviewed the annotation, split and evaluation records used to explain the research results. * I separated measured findings from planned improvements and system limitations.
04-10-2026	* I worked on the individual thesis draft using the project guide and saved evidence. * I described the dataset preparation, model development and application implementation in clear English.

October 2026

Research work record

05-10-2026	* I reviewed the prepared thesis draft, figures, references and appendices. * I checked that the reported component matched the developed counting and cycle-time workflow.
06-10-2026	* I revised the report wording and reviewed the remaining comments in the draft. * I kept the individual contribution separate from the other research components.
07-10-2026	* I checked consistency between the model description and the desktop application's behaviour. * I reviewed operation selection, session handling and production-history explanations.
08-10-2026	* I prepared demonstration notes covering device setup, session creation and live monitoring. * I included the recorded-video option for explaining the system without a sewing machine at the venue.

October 2026

Research work record

09-10-2026	* I organised the final component evidence and reviewed the development sequence for the viva. * I prepared explanations for the model choice, two-state rule, cycle-time logic and workstation detection.
10-10-2026	* I consolidated the component's development and documentation progress up to this date. * I recorded the main limitations and the remaining review points for the final thesis and demonstration.